



The Discordant

Codebase and design documentation

Code as of commit bef1089 · 2026-09-25 · private repository
JecerSE/the-discordant

Godot 4.7.2 · GDScript · 125 scripts · 10618 lines

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1. Overview

The Discordant is a 2D rhythm-action roguelite platformer. A quarter note falls off the Grand Score into the Margin and climbs back through three pillar bars (Percussion, Wind, Strings) to the Conductor. Every room is a page of sheet music: the five staff lines are the platforms. The Rest, a corruption of silence, muffles the music; attacking on the beat is stronger, and clearing a room brings the colour and the song back.

Technology

Item	Value
Engine	Godot 4.7.2 (stable), GL Compatibility renderer
Language	GDScript, statically typed where the analyzer allows
Resolution	1280 × 720 viewport, stretch mode canvas_items, aspect keep_height
Physics	60 ticks per second (engine default)
Art	None on disk. Everything is drawn in code with <code>_draw()</code> (Glyph and per-class drawing).
Audio	None on disk. Every sound is synthesised at startup in about 150 ms (Synth and src/audio).
Scenes	One <code>.tscn</code> (src/main.tscn). All other nodes are created in code.
Data	Game content in src/data/content/*.gd constants; tuning and combat data in .tres Resources under content/
Exports	Windows and Linux presets (export_presets.cfg), pck embedded, output in build/ (ignored by git)

Size

Folder	Scripts	Lines
src	1	54
src/actors/enemies	8	1095
src/actors/enemies/bosses	6	612
src/actors/player	10	1310
src/actors/player/powers	5	365
src/audio	4	374
src/autoload	3	528
src/autoload/game	2	242
src/data	4	491
src/data/combat	4	56
src/data/content	8	451
src/data/tuning	7	266
src/debug	2	138
src/fx	2	163
src/fx/effects	14	436
src/input	1	160
src/ui	13	1336
src/ui/hud	13	472
src/ui/map	2	185
src/world	5	779
src/world/generation	6	301
src/world/room	5	804

No script is longer than 300 lines (the largest is `src/actors/player/player_damage.gd` at 287).

Running

```
godot --path .                # play
godot --path . -e             # open in the editor
godot --headless --path . --script tools/smoke.gd # every room, boss, power and item with a bot
```

2. Game design

World and story

Element	In the game
The Grand Score	Every song there is. The Conductor writes it and never noticed the notes are alive.
The Margin	The hub and tutorial. Signs, a practice stand, Pause (a half rest who explains why rests are feared) and statues to choose a note.
The Rest	The corruption. Enemies are rests. While a room is hushed the music is low-passed and the melody drops out.
Pillars	Percussion (the Strikers, caverns, tradition), Wind (the Breathers, open sky, prejudiced against rests), String (the Resonants, humming forest, once played with the rests).
The Grand Score (area)	A short final bar: a fermata or shop, then the Conductor.
Secret	Three clefs hidden by the Scribble, one per pillar bar. Holding all three when the Conductor falls opens the fight with the Score itself.
Endings	Tacet (death), Prima volta (Conductor), Coda (the Score).

Characters

Note	Role	HP	Spd	Jmp	Dmg	DR	Starts with	Innate	Unlock
Quarter Note	all-rounder	100	270	2	×1.0	0.0	soundwave	Common Time: your beat window is 30% wider.	Default
Mundo	Whole Note · tank	170	215	1	×1.1	0.2	shockwave	Held: 20% less damage, no knockback, air attacks pound the ground. Percussion runes add +8% damage each.	Defeat the keeper of Percussion.
Half Note	balanced	120	250	2	×1.0	0.05	reverb_shield	Sustain: every melee hit rings again half a beat later for 45%.	Defeat the keeper of Strings.
Eighth Note	glass cannon	65	330	3	×1.2	0.0	gale_dash	Flag: three jumps, a dash that cuts, and damage that scales with speed.	Defeat the keeper of Wind.

Powers (20)

Two slots, a third with the Double Bar rune. Rehearsing at a fermata raises a power's level (up to III): +30% damage and -12% cooldown per level. Percussion scales with max HP, Wind with speed, String with how many runes you own.

Power	Family	Cooldown	Damage	What it does
Shockwave	Percussion	5.0 s	26	Slam the ground; a wave runs both ways along it. In the air, dive first.
Rimshot Parry	Percussion	2.2 s	20	A brief guard. Blocked hits are negated, shots are reflected, and your next strike lands on the beat.
Drumroll	Percussion	6.0 s	9	Six rapid strikes in front of you, each one counted on the beat.
Timpani Pillar	Percussion	7.0 s	22	Raise a pillar of stone ahead. It launches whatever stands there and stays as a wall for a while.
Gale Dash	Wind	3.0 s	18	A long dash that cuts through enemies. A kill refunds the cooldown.
Updraft	Wind	6.0 s	10	Fly up and hover for a moment. Hits enemies below you.
Breath Charge	Wind	4.0 s	34	Hold to charge, release to blow a gust. A full charge knocks enemies flying.
Fade	Wind	11.0 s	0	Turn invisible for 3 seconds. Your first hit out of it deals double.

Power	Family	Cooldown	Damage	What it does
Tempest	Wind	18.0 s	8	Summon a tornado that drags enemies in and lifts them, grinding them for 4 seconds.
Soundwave	String	1.1 s	16	Fire a wave of sound that passes through its target.
Echo	String	9.0 s	0	For 4 seconds, all damage you deal repeats one beat later.
Soul Pull	String	4.0 s	12	Hook the nearest enemy you face and reel it in, stunned.
Tether	String	9.0 s	0	Link up to four nearby enemies: damage to one is shared, 60%, with the rest.
Reverb Shield	String	8.0 s	0	A shield that absorbs 35 damage for 3 seconds, then releases what it caught as a ring.
Fermata	Rest	20.0 s	0	Everything but you stops for 3 seconds. Damage you deal is saved up and lands all at once, x1.5, when time starts again.
Caesura	Rest	12.0 s	30	Untouchable for 2 seconds but unable to strike. When it ends, everything near you is silenced.
Da Capo	Rest	15.0 s	0	Go back to where you were, and the health you had, 3 seconds ago.
Grace Note	Rest	2.5 s	20	Teleport behind the nearest enemy and hit it.
Accelerando	Rest	16.0 s	0	The whole bar plays at double tempo for 5 seconds. Enemies too.
Ghost Note	Rest	10.0 s	40	Leave a decoy of yourself. Enemies go after it. It explodes after 3 seconds.

Runes

Kind	Count	How it works	Runes
swap	10	Equip in the rune slots (3, or 4 with Segno). Swap outside combat.	Metronome Rune, Forte, Leggiero, Staccato, Legato, Crescendo, Accent, Dynamics, Theremin, Hurdy-Gurdy
family	12	Instrument runes. Two or three of one pillar wake a set bonus. Rival pillars together cause dissonance (a smaller beat window, and a combo effect).	Bass Drum, Cymbal, Snare Drum, Piccolo, Oboe, Trumpet, Flute, Violin, Guitar, Harp, Timpani, Piano
pause	5	Support runes about timing, parry and recovery.	Whole Rest, Half Rest, Quarter Rest, Eighth Rest, Fermata
permanent	9	Given by keepers. Locked for the run. Includes the form changers.	Key Signature, 4/4, Coda, Double Bar, Repeat Sign, Segno, Diminution, Augmentation, Dotted Rhythm
margin	6	Rule-breaking and risky. Mostly from the Scribble.	Tritone, Glass Harmonica, Out of Tune, 4'33", Syncopation, Kazoo

Relics (18) and champion items (6)

Relic	Price	Effect
Rosin	70 #	+12% damage.
Tuning Fork	80 #	On-beat hits heal 1 HP.
Brass Metronome	60 #	Beat window +30 ms.
Sheet Protector	65 #	+20 max HP.
Ear Plugs	75 #	Take 12% less damage.
Valve Oil	60 #	+10% move speed, dash cooldown 15% shorter.
Capo	75 #	Power cooldowns 12% shorter.
Page Turner	70 #	+1 jump.
Practice Mute	55 #	Enemy projectiles fly 25% slower.
Encore Ticket	60 #	Heal 8 HP when a room is cleared.
Gold Leaf	50 #	+35% sharps.

Relic	Price	Effect
Espresso	70 #	+15% attack speed. Shaky hands.
Sustain Pedal	80 #	Powers deal +22% damage.
Tempo Marking	55 #	Your first attack in each room always lands on the beat.
Bent Paperclip	30 #	+1 sharp per kill. Found it on the floor.
Pencil Stub	85 #	Heal for 3% of the damage you deal.
Double Stop	90 #	Melee hits strike a second time for 30%.
Rubber Mallet	60 #	Melee knocks enemies back much farther.

Champion item	Family	Effect
Timpanist's Mallet	Percussion	Every fourth melee hit releases a shockwave.
Cymbal Shard	Percussion	On-beat hits crash in a ring for 45%. +10 max HP.
Piper's Reed	Wind	Dashing through enemies cuts them for 20. Dash cooldown 25% shorter.
Hornist's Bell	Wind	+20% move speed. Damage scales with speed.
Violist's Bow	String	Every melee hit also fires a small soundwave.
Cellist's Endpin	String	+40 max HP. Taking a hit rings out for 15 damage around you.

Enemies (23)

Every enemy acts on the beat: a red accent mark (>) appears one beat before it strikes. Health and damage grow per bar ($\times 1.45$ and $\times 1.22$ per bar index).

Enemy	AI	HP	Dmg	Trick
Quarter Rest	walker	30	10	
Whole Rest	dropper	46	16	
Snare Rest	jumper	34	12	
Half Rest	charger	52	14	
Eighth Rest	flyer	24	10	
Sixteenth Rest	shooter	22	9	
Gust Rest	gust	38	8	
Tether Rest	tether	42	4	
Echo Rest	echo	32	11	
Rimshot Guard	guard	48	12	Its stance turns aside anything off the beat. Strike it ON the beat to shatter it.
Bandleader Rest	buffer	40	8	Rallies the rests around it: faster, harder. Kill it first.
Staccato Rest	dasher	28	13	Winds up, then bursts across the room. Interrupt it or get out of the line.
Breathless Rest	phantom	30	10	Fades out of sight. Only area attacks can hit it, or stun it to make it show up.
Breath Reed	well	70	0	Breathes for every wind rest in the room. Break it and they all choke.
Motif Rest	motif	30	8	Its waves mark you. Three marks and it goes off on you.
Unison Rest	binder	44	6	Binds itself to you. While bound, every hit you land on it lands on you instead. Wait it out.
Reverb Warden	warden	50	10	A pulsing barrier throws shots back and staggers whoever strikes it. Hit it between pulses.
Fallen Timpanist	elite_timpanist	300	18	elite
Fallen Cymbalist	elite_cymbalist	270	16	elite
Fallen Piper	elite_piper	240	14	elite
Fallen Hornist	elite_hornist	280	16	elite

Enemy	AI	HP	Dmg	Trick
Fallen Violist	elite_violist	260	13	elite
Fallen Cellist	elite_cellist	360	18	elite

Bosses

Boss	Title	HP	Family
The Hollow Timpani	keeper of Percussion	1000	Percussion
The Breathless Flute	keeper of Wind	950	Wind
The Unstrung Harp	keeper of Strings	1100	String
The Conductor	who never noticed the notes were alive	2100	Podium
The Score	the sheet itself, awake	3000	Grand

Bars (areas) and tempos

Area	Subtitle	BPM	Boss
The Margin	outside the staff, where erased notes end up	92.0	
Bar I: Percussion	the Strikers, deep in the caverns	108.0	timpani
Bar II: Wind	the Breathers, up in the open sky	124.0	flute
Bar III: Strings	the Resonants, in the humming forest	96.0	harp
The Grand Score	the top of the world	116.0	conductor
The Score, Itself	the page under all the pages	132.0	grand_staff

3. Architecture

The prototype is a code-first Godot project: one scene, nodes built with `.new()`, data in constants and Resources, drawing in code. Large classes are split into chains of layer classes so no file passes 300 lines. Each layer extends the one below; the top layer keeps the class name everything else uses.

Class layer chains

Class	Layers, bottom to top	Rule
Player	PlayerState → PlayerDamage → PlayerAttacks → PlayerHooks → PlayerMovement → Player	A layer may only call functions in its own file or below.
Enemy	EnemyState → EnemyAttacks → EnemyDamage → EnemyMoveAI → EnemyBeatAI → EnemyEliteAI → Enemy	Bosses extend Enemy through Boss.
Room	RoomState → RoomCombat → RoomFlow → Room	Lower layers pass themselves as Room with a cast (self as Room).
Game (autoload)	GameCore → GameRun → game.gd	The autoload node is game.gd.
Content	Front door aliasing CharactersData, PowersData, RunesData, RelicsData, EnemiesData, PagesData, StoryData	Content.X still works everywhere.
Powers	Front door dispatching to PercussionPowers, WindPowers, StringPowers, RestPowers	Shared helpers stay on Powers.
FX	Front door of preload constants for 13 effect classes (FxRing, FxSlash, ...)	FX.Ring.new() still works.

Autoloads

Autoload	Role	Notes
Beat	The clock: step (sixteenth), beat and bar signals; offsets for judging presses; song position in beats.	Runs while paused (process mode ALWAYS). Every beat listener checks <code>get_tree().paused</code> .
Synth	Builds all samples at startup, plays them from a pool of 28 players, runs the generative song, muffles and crossfades music.	Music and SFX buses are created at runtime.
Game	Settings, save file, meta progress, the current run, derived stats, input install, screen routing, god mode flag.	Tools launched with <code>--script</code> use a separate test save.

Screen flow

Screen	Script	Leads to
Title	src/ui/title.gd	Begin → the Margin; Prologue; Settings; Quit
The Margin (hub)	Room type hub	Walk into the exit → a new run → Map
Map	src/ui/map_screen.gd + src/ui/map/	Pick a note → Room
Room	src/world/room/	Exit → Map; boss → next bar's Map or an ending; death → Ending
Ending	src/ui/ending.gd	Confirm → the Margin

Screens change through `Game.goto()`, then a deferred swap in `main.gd` that unpauses, frees the old screen, adds the new one and fades from paper colour.

Runtime node tree of a room

Room (Node2D)	
StaticBody2D (layer 1)	floor, walls, ceiling
StaticBody2D x N (layer 2)	one per staff-line segment, one-way
Background (Node2D, z -10)	paper, staff, wash, segments, features
Node2D actors	Player (+ Camera2D, MeleeSwing children), Enemy, Boss
Node2D projectiles	Projectile
Node2D fx	FX effects
Interactable x N	chests, shop items, teachers, bench, statues, signs

Hud (CanvasLayer 5)
Timer x N

one widget per file + the open overlay
delayed callbacks (Room._after)

Collision layers

Layer	Bit value	Used by
1 solid	1	Floor, walls, ceiling, Timpani pillars
2 one-way lines	2	Staff-line segments (the player drops through with Down + Jump)
3 player	4	Player body
4 enemies	8	Enemy bodies (flyers only collide with layer 1)

Hits are distance and rectangle checks against `room.alive_enemies()`, not `Area2D`. This keeps them deterministic and cheap.

Communication

Emitter	Signal	Listener
Beat	<code>step(n)</code>	Synth sequencer
Beat	<code>beat(n)</code>	<code>Enemy._on_beat</code> → <code>ai_beat</code> , <code>Player._on_beat</code> (Theremin, Hurdy-Gurdy), <code>HudMetronome</code>
Beat	<code>bar(n)</code>	<code>Player._on_bar</code> (String set auto-wave)
Game	<code>toast(text, color)</code>	<code>HudToasts</code>
MeleeSwing	<code>struck(enemy, damage, info)</code> , <code>landed_first_hit</code>	Player (deals damage, pogo bounce)
Projectiles, FX, enemies	(direct calls, no signal)	Call <code>player.deal()</code> and <code>player.take_hit()</code> through <code>room.player</code>
Overlays	Callable fields (offer, dialogue, menu)	Room and Events callbacks

The prototype still relies on direct calls through `room` (a god object) far more than on signals. `ARCHITECTURE.md` Part 2 sets the rules for the rebuild: components, exported references, call down / signal up, an event bus, a headless core and a command queue.

4. Systems

Rhythm and timing grades

Every attack or power press is graded the moment it is pressed, against the nearest beat (half-beat for the Eighth Note). `Beat.signed_offset()` subtracts the audio output latency and the player's beat offset setting.

Grade	Window	Damage multiplier	Counts as on the beat
Perfect	≤ 0.03 s	1.65 + rune beat bonus	Yes
Great	$\leq$ the player's beat window (0.085 s base, $\times 1.3$ for Quarter)	1.45 + rune beat bonus	Yes
Good	$\leq$ beat window $\times 1.8$	1.15	No
Miss	anything further	0.85, minus 0.05 per stacked miss (max 6 stacks)	No
None	hits without a press (echoes, follow-ups)	1.0	No

Rhythm combos

ComboTracker remembers presses in beats. When the gaps between the latest presses match a pattern (within ± 0.2 beat) and every press was at least good, the finisher replaces that attack. All great or better makes it perfect (damage $\times 1.5$).

Note	Combo	Rhythm	Gaps (beats)	Dmg	Finisher
Quarter	Common Time	♪ ♪ ♪ ♪	(1, 1, 1)	34.0	Four quarters in a row: a wide crescent plus a soundwave.
Quarter	Syncopated Run	♪ ♪ ♪	(0.5, 0.5)	24.0	Two eighths then a quarter: dash through everything ahead.
Half	Cut Time	half half half	(2, 2)	40.0	Three halves: two rings, the second one beat after the first.
Half	Dotted Rise	♪. ♪ ♪	(1.5, 0.5)	30.0	Dotted quarter, eighth, quarter: a rising slash that launches enemies.
Whole	Semibreve	whole whole	(4)	48.0	Two strikes a whole bar apart, nothing between: a quake that stuns.
Whole	Stomp Time	♪ ♪ ♪	(1, 1)	30.0	Three quarters: a heavy ring that stuns everything near.
Eighth	Eighth Run	♪ ♪ ♪ ♪ ♪	(0.5, 0.5, 0.5, 0.5)	9.0	Five eighths in a row: a flurry of six quick cuts.
Eighth	Swing Cut	♪ ♪ ♪	(0.5, 1)	26.0	Eighth, quarter, eighth: blink behind the nearest enemy and cut.

Attacks

Each character's chain is an `AttackSet` resource (`content/combat/*_attacks.tres`). Every swing spawns a `MeleeSwing`, a child of the player, so the hitbox moves with the player for its active time and hits each enemy once. Attacking in the air while holding Down uses the down strike; if it connects the player bounces up (pogo). Mundo's air attack stays a ground pound.

Note	Chain damage	Cadence (s)	Down strike	Graded on
Quarter	10.0, 10.0, 17.0	0.26, 0.26, 0.34	12.0	beat
Half	16.0, 22.0	0.4, 0.4	19.0	beat
Whole	24.0	0.56	ground pound	beat
Eighth	7.0, 7.0, 7.0, 12.0	0.16, 0.16, 0.16, 0.16	9.0	half-beat

Damage pipeline

```
mult = base_dmg * (1 + dmg)
      * (1 + power_dmg)      powers
      * (1 + proj_dmg)       projectiles
      * speed scaling        Hornist's Bell / Wind set 3
      * grade multiplier     perfect / great / good / miss (see above)
      * crescendo, accent, stunned bonus, Fade, Out of Tune, Diminution
amount = base * mult
then: lifesteal, on-beat heal, cooldown refunds, Kazoo, Tether share, Echo,
      Sustain (Half Note), Double Stop, Cymbal crash, Pizzicato, Mallet shockwave
```

All of this lives in `PlayerDamage.deal()` (`src/actors/player/player_damage.gd`). Follow-up hits pass `proc: false` so they never chain. Damage numbers go through `DamageNumbers`, which merges hits on the same target within 220 ms into one number.

Movement

Value	Setting	Where
Gravity / jump speed / max fall	2100 / 790 / 980 px/s	PlayerState constants
Jump height	$790^2 / (2 \times 2100) \approx 149$ px (staff lines are 110 px apart)	derived
Coyote time / input buffer	0.10 s / 0.13 s	PlayerState constants
Dash	720 px/s for 0.15 s, 0.75 s cooldown, untouchable	PlayerState constants
Jump while rising fast	adds 0.35 of jump speed instead of replacing the rise	PlayerMovementTuning
Momentum carry	air control $\times$ 0.35 for 0.35 s after a launch or strong push	PlayerMovementTuning
On-beat jump / dash / landing	+0.12 jump, +0.35 dash time, +0.15 speed for 0.5 s	PlayerMovementTuning
Drum pad / updraft	launch 1180 px/s / lift 3400 px/s ² up to 560 px/s	PlayerMovementTuning

Enemies and bosses

- Each enemy runs `ai_process()` every frame (movement) and `ai_beat()` on every beat (attacks). A wind-up shows the accent mark one beat ahead.
- Detection: an enemy notices the player within 560 px and gives up past 900 px; elites and bosses always know. Taking damage alerts it.
- Idle enemies patrol random points within 260 px of their spawn, pausing 0.4 to 1.4 s, with a 0.35 s turn cooldown.
- Elites and bosses have super armor while winding up or attacking, and can't be stunned again for 2.0 s after a stun. Bosses take 0.2 of any stun.
- Bosses phrase attacks in bars and change pattern at health thresholds. The Hollow Timpani uses `TimpaniTuning` (a 2-bar cycle with a recovery window).

Generation

Step	What happens	Class
Bar layout	Entry fight, branching layers of 2, 3, 3, 3, 2 rooms, then a fermata (sometimes with a chest), then the boss. Specials placed once each: elite, shop, chest, chest, teach, teach, plus a second elite 60% of the time.	BarPlanner, MapGen
Room size	Fights 2300 to 3200 px wide (+200 on Wind), elites 1900 px, events 1280 px.	Layout
Platforms	Each staff line is broken into segments by per-pillar densities.	Layout
Reachability	A search from the floor, one line up at a time within 110 px (same line 150 px). Unreachable platforms get a stepping-stone ledge below.	PlatformReachability
Features	Drums, updrafts and harmonic nodes at least 280 px apart, clear of spawn and exit.	FeaturePlacer
Waves	2 to 4 waves, varied sizes, at most 2 of one type, reinforcements 35% of the time once half a wave is down.	WavePlanner, WavePlan

Step	What happens	Class
Spawns	Floor, platforms (weighted by length) and air, 170 px apart and 320 px from the player.	SpawnPicker
Offers	Items that need a skill you lack are skipped; offers lean to your pillars.	Loot, ItemRequirements

All generation uses the run's seeded RandomNumberGenerator, so a seed always gives the same map and rooms.

Music

- Each area has a seeded generative song: drums, bass, pad, arpeggio and a lead phrase in an A B A' C shape.
- While a room is hushed the Music bus is low-passed (18000 Hz → 520 Hz) and the lead is muted above half hush.
- The muffle follows how much of the room is left (never below the minimum while enemies remain) and clears on room clear.
- Changing bars crossfades the songs (MusicTuning). The tempo per area is fixed; revisiting it waits for the real music (issue #6).

Interface

Part	Files	Notes
HUD	src/ui/hud/ (Hud + 11 widgets)	Health, power bar (under health), metronome, boss bar, room status, edge flash, banner, toasts, prompt with item details, character card
Map	src/ui/map_screen.gd, src/ui/map/	Drawn as a score: notes per room type, measures, slurs, playhead
Overlays	src/ui/*.gd extending Overlay	Choice cards, dialogue, menu list, loadout, pause, settings (sliders), controls (rebinding)
Title / Ending	src/ui/title.gd, ending.gd	Prologue, stats, endings

Input and rebinding

InputBindings (src/input/input_bindings.gd) registers 13 actions with defaults, then applies the player's saved bindings from settings. The Controls menu shows three keyboard/mouse slots and one controller slot per action; binding an input removes it from any other action. InputLabels turns bindings into short names (LMB, RB, D-pad up) for on-screen hints.

Action	Defaults (keys, mouse:N, pad:N, axis)
Move left	A, LEFT, pad:13, axis:0:-1
Move right	D, RIGHT, pad:14, axis:0:1
Up / talk	W, UP, pad:11, axis:1:-1
Down	S, DOWN, pad:12, axis:1:1
Jump	SPACE, Z, pad:0
Attack	J, X, mouse:1, pad:2
Power 1	K, C, mouse:2, pad:3
Power 2	L, V, pad:1
Power 3	U, B, pad:9
Dash	SHIFT, I, pad:10
Use / talk	E, F, pad:7
Runes	TAB, Q, pad:4
Pause	ESCAPE, P, pad:6

Saves, settings and debug

- Save: user://discordant_save.json with meta (unlocks, runs, wins, seen enemies) and settings (volumes, shake, beat offset, metronome, fullscreen, bindings).
- Anything launched with --script uses user://discordant_test_save.json, so tests never touch the real save.
- Debug menu (F1): god mode, heal, sharps, grant items, learn a power, kill / clear, clefs, unlock characters, skip to next bar or boss. Available in debug builds, or with -- --debug-menu.

5. Data and tuning

Content (what exists) lives in GDScript constant files; tuning (how it behaves) lives in Resource scripts with .tres instances under content/, editable in the inspector. New numbers must go in a tuning resource, never inline.

Content file	Holds
src/data/content/characters_data.gd	The four notes
src/data/content/powers_data.gd	20 powers
src/data/content/runes_data.gd	Runes, family sets, dissonance
src/data/content/relics_data.gd	Relics and champion items
src/data/content/enemies_data.gd	Enemies and bosses
src/data/content/pages_data.gd	Areas, songs, the climb, clefs
src/data/content/story_data.gd	Teachers, intro, prologue, endings
src/data/content/item_requirements.gd	Which items need which skill
content/combat/*_attacks.tres, *_combos.tres	Attack chains and rhythm combos per character

EnemyTuning (src/data/tuning/enemy_tuning.gd)

Tuning for how regular enemies notice and wander. Edit content/tuning/enemy_tuning.tres.

Field	Default	Meaning
aggro_range	560	An enemy notices the player inside this distance (px).
leash_range	900	...and gives up once the player is farther than this.
patrol_radius	260	How far from its spawn point an idle enemy wanders (px).
patrol_min_step	80	New patrol points are at least this far from the current position (px).
patrol_arrive_distance	14	Distance at which a patrol point counts as reached (px).
patrol_pause_min	0.4	Shortest pause at a patrol point (s).
patrol_pause_max	1.4	Longest pause at a patrol point (s).
turn_cooldown	0.35	Minimum time between turning around at a wall or ledge (s).
patrol_speed_scale	0.5	Patrol speed as a fraction of the enemy's chase speed.
boss_stun_scale	0.2	Bosses take this fraction of any stun's duration.
stun_immunity	2	After a stun ends, elites and bosses can't be stunned again for this long (s).

FeedbackTuning (src/data/tuning/feedback_tuning.gd)

Tuning for on-screen combat feedback. Edit content/tuning/feedback_tuning.tres.

Field	Default	Meaning
damage_merge_window_ms	220	Hits on the same target within this window add into one number instead of stacking.
damage_font_size	18	Size of a normal damage number.
on_beat_font_size	22	Size of an on-the-beat damage number.
channel_font_size	16	Size of echo, shared and stored numbers.
merge_pop_size	6	Extra size added briefly when a merged hit bumps an existing number.

LevelGenTuning (src/data/tuning/level_gen_tuning.gd)

Room and map generation. Edit content/tuning/level_gen_tuning.tres.

Field	Default	Meaning
combat_width_min	2300	Narrowest fight room (px).
combat_width_extra	900	Up to this much wider, at random (px).
wind_width_bonus	200	Extra width for Wind fight rooms (open sky) (px).
elite_width	1900	Width of an elite room (px).
segment_length_min	200	Shortest staff-line platform (px).
segment_length_max	560	Longest staff-line platform (px).
gap_min	120	Smallest gap between platforms on a line (px).
gap_max	280	Largest gap between platforms on a line (px).
gap_per_line	20	Extra gap per line up, so higher lines are sparser (px).
start_margin	220	Leave this much of the left edge (spawn) and right edge (exit) free of line ink.
end_margin	160	Keep this much of the right edge free of platforms (px).
climb_reach	110	Horizontal edge-to-edge distance a note can cover while climbing one line (px). Based on Mundo, the note with one jump and the lowest speed.
same_line_reach	150	Gap a note can clear jumping across the same line (px).
stepping_stone_length	180	Length of the ledge added under a platform nobody can reach (px).
feature_min_spacing	280	Minimum distance between any two floor features (drums, updrafts, harmonics) (px).
feature_spawn_clearance	420	Keep features this far from the spawn (left) and the exit (right) (px).
feature_exit_clearance	380	Keep features this far from the exit (px).
feature_tries	24	Placement attempts per feature before giving up.
waves_min	2	Fewest waves in a fight room.
waves_max	4	Most waves in a fight room (also limited by the bar).
wave_size_min	3	Smallest wave.
wave_size_extra	2	Up to this many more enemies, at random.
wave_size_per_bar	1	Extra enemies per wave for each bar climbed.
wave_size_cap	8	Largest possible wave.
max_same_type_per_wave	2	No enemy type appears more than this many times in one wave.
reinforcement_chance	0.35	Chance that a wave calls reinforcements once half of it is down.
reinforcement_size_min	1	Fewest reinforcements.
reinforcement_size_max	3	Most reinforcements.
spawn_spacing	170	Enemies in one wave spawn at least this far apart (px).
spawn_player_distance	320	...and at least this far from the player (px).
branch_layers	2, 3, 3, 3, 3, 2	Rooms per branching layer between the entry fight and the fermata.
special_rooms	elite, shop, chest, chest, teach, teach	Special rooms placed once each across the branching layers.
second_elite_chance	0.6	Chance a bar gets a second elite on its last branching layer.
rest_or_chest_chance	0.35	Chance the fermata layer also offers a chest.
bonus_drop_chance	0.4	Chance that a cleared fight room offers a bonus pick.

MusicTuning (src/data/tuning/music_tuning.gd)

How the music reacts to play (issue #13). Edit content/tuning/music_tuning.tres.

Field	Default	Meaning
min_combat_hush	0.2	While a room still has rests in it, the music never clears past this much muffle.
hush_follow_speed	0.8	How fast the muffle follows the room (hush units per second).
crossfade_time	0.35	Fade out, switch songs, fade in when moving between bars (s, each way).
crossfade_depth_db	-30	How far the music dips during a crossfade (dB).

PlayerMovementTuning (src/data/tuning/player_movement_tuning.gd)

Player movement values added after the prototype (momentum, launchers, on-beat)

Field	Default	Meaning
jump_stack_ratio	0.35	A jump while already rising faster than jump speed adds this fraction of jump speed instead of replacing the rise.
momentum_carry_time	0.35	After a launch or a strong push, air steering is reduced for this long (s)...
momentum_air_control	0.35	...to this fraction of normal air acceleration, so the momentum carries.
momentum_push_threshold	200	Horizontal pushes stronger than this (px/s) start a momentum carry.
launch_lock_time	0.08	After a launch, the grounded reset (jumps, coyote) is skipped for this long (s).
beat_jump_bonus	0.12	A jump pressed on the beat goes this much higher (fraction of jump speed).
beat_dash_bonus	0.35	A dash pressed on the beat lasts this much longer (fraction).
flow_time	0.5	Landing on the beat gives a short burst of speed ("flow").
flow_speed_bonus	0.15	Speed bonus during flow (fraction).
drum_launch_speed	1180	Upward speed a drum pad launches you at (px/s).
updraft_accel	3400	Upward acceleration inside an updraft (px/s ²).
updraft_max_rise	560	Fastest rise inside an updraft (px/s).

TimingTuning (src/data/tuning/timing_tuning.gd)

How attack timing is graded and rewarded (issues #12 and #14).

Field	Default	Meaning
perfect_window	0.03	Within this of the beat (s) is "perfect". "Great" uses the player's beat window (base 85 ms, wider with runes). "Good" is the beat window times good_window_scale.
good_window_scale	1.8	"Good" window as a multiple of the beat window.
perfect_multiplier	1.65	Damage for a perfect (before rune beat bonuses).
great_multiplier	1.45	Damage for a great (before rune beat bonuses).
good_multiplier	1.15	Damage for a good.
miss_multiplier	0.85	Damage for a miss (before the mash penalty).
mash_penalty_step	0.05	Each consecutive missed attack takes this much more off a miss (fraction)...
mash_penalty_max_stacks	6	...up to this many stacks. A good-or-better hit clears the stacks.
combo_interval_tolerance	0.2	A gap between presses may be off its written length by this many beats.
combo_memory_beats	9	Presses older than this many beats are forgotten.
combo_perfect_multiplier	1.5	Finisher damage multiplier when every press was great or better.

TimpaniTuning (src/data/tuning/timpani_tuning.gd)

Pattern and speed for the Hollow Timpani (issue #27: slowed down).

Field	Default	Meaning
cycle_beats	8	Beats in one attack cycle for phases 1-2 and for phase 3.
cycle_beats_phase3	4	Beats in one attack cycle in phase 3 (faster).
windup_beat	0	Beat in the cycle that shows the wind-up; the leap comes one beat later.
mallet_beat	6	Beat in the cycle when mallets are thrown (phases 1-2 only).
leap_air_beats	1	Airtime of a leap, in beats. It lands on a beat, then stands still until the next wind-up.
leap_max_speed	620	Fastest horizontal speed of a leap (px/s).
wave_speed	380	Landing shockwave speed in phase 1 (px/s).
wave_speed_per_phase	40	Extra shockwave speed per phase (px/s).
wave_life	2.2	How long a shockwave travels (s).
wave_height	36	Shockwave height; jump over it (px).
landing_ring_radius	130	Radius of the ring around a landing (px).
mallets_phase1	2	Mallets thrown per volley in phase 1.
mallets_phase2	3	Mallets thrown per volley in phase 2.
mallet_speed_scale	0.75	Mallet throw speed as a fraction of the original.
mallet_lift	620	Upward speed of a thrown mallet (px/s).
mallet_gravity	1100	Gravity on thrown mallets (px/s ²).
mallet_damage_scale	0.7	Mallet damage as a fraction of the boss damage.
summon_every_beats	16	From phase 2, summon Snare Rests this often (beats).
summon_count	2	Snare Rests per summon.
summon_max_alive	3	No summon while this many other enemies are alive.

6. File map

Every script in `src/`, with its class and the first line of its description.

File (under <code>src/</code>)	Lines	Class	Extends	Description
<code>actors/enemies/bosses/boss.gd</code>	113	Boss	Enemy	A keeper, or worse. Bosses phrase their attacks in bars: they wind up on one beat and strike on the next, and every phase changes the phrase.
<code>actors/enemies/bosses/conductor.gd</code>	155		Boss	The Conductor, author of all reality, who never knew the notes were alive. The fight is a symphony in movements: Allegro (percussion), Adagio (strings), Presto (wind), and a Finale where the tempo climbs and everything plays at...
<code>actors/enemies/bosses/flute.gd</code>	86		Boss	The Breathless Flute, keeper of Wind, holding its breath. It drifts across the sky, breathes notes at you on the beat, and exhales gales that shove you across the page.
<code>actors/enemies/bosses/grand_staff.gd</code>	70		Boss	The Score, itself, the sheet every note lives on, woken by three clefs. The arena is its body: its lines strike on the beat, notes rain from its margins, and a great clef-eye drifts between the lines. Strike the eye.
<code>actors/enemies/bosses/harp.gd</code>	100		Boss	The Unstrung Harp, keeper of Strings, out of tune. Its strings run the height of the page: each one glows on a beat and snaps on the next. It pulls you close to play you.
<code>actors/enemies/bosses/timpani.gd</code>	88		Boss	The Hollow Timpani, keeper of Percussion. It leaps on the beat and lands on the next; the landing runs along the floor. Jump the waves.
<code>actors/enemies/enemy.gd</code>	203	Enemy	EnemyEliteAI	A Rest, a soldier of the Tacet, the silence eating the Grand Staff. Every enemy acts on the beat: it winds up (an accent mark appears over it) on one beat and strikes on the next, so a player who is listening always gets a...
<code>actors/enemies/enemy_attacks.gd</code>	85	EnemyAttacks	EnemyState	Layer 2 of 7. Shots, shockwaves, rings, tethers and binds.
<code>actors/enemies/enemy_beat_ai.gd</code>	172	EnemyBeatAI	EnemyMoveAI	Layer 5 of 7. On-beat behavior for the regular Rests (wind up, then strike).
<code>actors/enemies/enemy_damage.gd</code>	111	EnemyDamage	EnemyAttacks	Layer 3 of 7. Taking damage, stuns, pushes, Fermata release, death.
<code>actors/enemies/enemy_elite_ai.gd</code>	104	EnemyEliteAI	EnemyBeatAI	Layer 6 of 7. On-beat behavior for the six fallen champions. <code>ai_beat</code> is the entry point: regular ai types are passed down to <code>_beat_basic</code> .
<code>actors/enemies/enemy_move_ai.gd</code>	206	EnemyMoveAI	EnemyDamage	Layer 4 of 7. Per-frame behavior for every ai type (movement, charges, swoops).
<code>actors/enemies/enemy_patrol.gd</code>	55	EnemyPatrol	RefCounted	Where an enemy wanders while it hasn't noticed the player: random points around its spawn, with a short pause at each. Pure logic, no Node; the enemy owns one.
<code>actors/enemies/enemy_state.gd</code>	159	EnemyState	CharacterBody2D	Layer 1 of 7. Every field an enemy has, setup from Content, targeting and small queries. Split from the original <code>enemy.gd</code> without logic changes.
<code>actors/player/beat_grader.gd</code>	50	BeatGrader		Grades how close a press landed to the beat, and what that grade is worth (issue #12). No state: the player keeps its own mash stacks.
<code>actors/player/combo_finishers.gd</code>	69	ComboFinishers		What each rhythm combo does when it completes (issue #11). Every finisher counts as on the beat; a perfect combo multiplies its damage.
<code>actors/player/combo_tracker.gd</code>	55	ComboTracker	RefCounted	Remembers recent attack presses (in beats) and reports when the latest ones form one of the character's rhythm patterns (issue #11).

File (under src/)	Lines	Class	Extends	Description
actors/player/melee_swing.gd	95	MeleeSwing	Node2D	A melee hitbox attached to the player (issue #10). As a child of the player it moves with every step, jump and fall for its active time, and strikes each enemy once. It never deals damage itself: it signals up and the player...
actors/player/player.gd	67	Player	PlayerMovement	The note you play as. Movement, the four characters' attacks, the damage pipeline every rune and relic hooks into, and the state the powers (powers.gd) toggle.
actors/player/player_attacks.gd	204	PlayerAttacks	PlayerDamage	Layer 3 of 6. Timing grades, the four characters' attacks (data in content/combat/), swings that follow the player, rhythm combos, dash.
actors/player/player_damage.gd	287	PlayerDamage	PlayerState	Layer 2 of 6. Damage out (deal and every rune hook on a hit) and damage in (take_hit, parry, shield, heal, death).
actors/player/player_hooks.gd	73	PlayerHooks	PlayerAttacks	Layer 4 of 6. Runes and effects that run on the clock: bar and beat hooks, 4'33", Da Capo history, Motif marks, stagger.
actors/player/player_movement.gd	286	PlayerMovement	PlayerHooks	Layer 5 of 6. Setup and the per-frame loop: input, run, jump, gravity, landing, timers.
actors/player/player_state.gd	124	PlayerState	CharacterBody2D	Layer 1 of 6. Every field the player has, plus small read-only helpers. Split from the original player.gd without logic changes.
actors/player/powers/percussion_powers.gd	39	PercussionPowers		Percussion powers: shockwave, parry, drumroll, pillar. They scale with max HP. Split from the original powers.gd; each case is unchanged.
actors/player/powers/powers.gd	127	Powers		Front door for active powers: cooldowns, levels, pillar scaling, casting, and the shared helpers (shockwaves, soundwaves). Each family's powers live beside this file. Levels (from rehearsing at a fermata) add 30% damage and cut...
actors/player/powers/rest_powers.gd	80	RestPowers		Rest powers, taught by the Scribble: fermata, caesura, da capo, grace note, accelerando, ghost note. Split from the original powers.gd; each case is unchanged.
actors/player/powers/string_powers.gd	52	StringPowers		String powers: soundwave, echo, soul pull, tether, reverb shield. They scale with rune count. Split from the original powers.gd; each case is unchanged.
actors/player/powers/wind_powers.gd	67	WindPowers		Wind powers: gale dash, updraft, breath charge, fade, tempest. They scale with speed. Split from the original powers.gd; each case is unchanged.
audio/synth_drums.gd	63	SynthDrums		Drum kit samples.
audio/synth_dsp.gd	38	SynthDsp		Shared helpers for sample generation: buffers, fades, WAV packing, MIDI to Hz.
audio/synth_instruments.gd	114	SynthInstruments		Pitched instruments, each generated once at one base note and pitched by pitch_scale.
audio/synth_sfx.gd	159	SynthSfx		Sound effects: hits, pickups, UI, bosses.
autoload/beat.gd	116		Node	The metronome everything obeys. Enemies strike on beats, bosses phrase their patterns in bars, the synth plays on sixteenths, and every attack the player makes is judged against the nearest beat.
autoload/game.gd	131		GameRun	Run state, meta progress, settings, input, and moving between screens. Tools (smoke tests, captures) run with --script and must never touch the real save.
autoload/game/game_core.gd	73	GameCore	Node	Layer 1 of 3 of the Game autoload. Signals, settings, meta progress, the save file and the input map. Split from the original game.gd without logic changes.
autoload/game/game_run.gd	169	GameRun	GameCore	Layer 2 of 3. The run: starting one, pages, nodes, keepers, items and powers.

File (under src/)	Lines	Class	Extends	Description
autoload/synth.gd	281		Node	Sample generation lives in src/audio (SynthDsp, SynthInstruments, SynthDrums, SynthSfx). Every sound in the game is synthesised here at startup: drums, four lead instruments, a bass, a pad, and the effects. Nothing is loaded from...
data/combat/attack_set.gd	14	AttackSet	Resource	A character's basic attack: the chain of swings, when the chain resets, and the downward strike used in the air while holding down (issue #10).
data/combat/attack_step.gd	22	AttackStep	Resource	One swing of a character's attack chain. Offsets and sizes are in px, relative to the player and mirrored by facing. The hitbox follows the player while active.
data/combat/combo_pattern.gd	15	ComboPattern	Resource	A rhythm combo (issue #11). The pattern is the gaps between attack presses, in beats: [1, 1, 1] is four presses on consecutive beats, [0.5, 0.5, 1] is two eighths and a quarter. Every press must be at least a "good"; all "great"...
data/combat/combo_set.gd	5	ComboSet	Resource	The rhythm combos one character knows.
data/content.gd	49	Content		The front door to all game data. The data itself lives in src/data/content/*.gd; logic elsewhere looks things up here by id.
data/content/characters_data.gd	40	CharactersData		The four playable notes. Split from the original content.gd without changing any values.
data/content/enemies_data.gd	49	EnemiesData		Enemies (the Rests), elites and bosses. Split from the original content.gd without changing any values.
data/content/item_requirements.gd	69	ItemRequirements		Which items only help if you already have a certain kind of skill (issue #26). Loot skips an item whose requirement you don't meet, so every offer does something for your current kit.
data/content/pages_data.gd	55	PagesData		Pages (areas), their songs, the climb order and the clefs. Split from the original content.gd without changing any values.
data/content/powers_data.gd	50	PowersData		Active powers: Percussion, Wind, String and Rest. Split from the original content.gd without changing any values.
data/content/relics_data.gd	32	RelicsData		Relics and champion memorabilia. Split from the original content.gd without changing any values.
data/content/runes_data.gd	81	RunesData		Runes, family set bonuses and dissonance combos. Split from the original content.gd without changing any values.
data/content/story_data.gd	75	StoryData		Teachers, the Margin intro, the prologue and the endings. Split from the original content.gd without changing any values.
data/glyph.gd	266	Glyph		Engraving by hand. Every note, rest, clef and accidental in the game is drawn by these functions onto whatever CanvasItem asks, so the whole look is one consistent ink.
data/loot.gd	111	Loot		Drawing items from the pools without repeats.
data/pal.gd	65	Pal		Colours and type. The world is ink on paper; each pillar paints the page its own wash once the Tacet is driven out of a room.
data/tuning/enemy_tuning.gd	31	EnemyTuning	Resource	Tuning for how regular enemies notice and wander. Edit content/tuning/enemy_tuning.tres.
data/tuning/feedback_tuning.gd	15	FeedbackTuning	Resource	Tuning for on-screen combat feedback. Edit content/tuning/feedback_tuning.tres.
data/tuning/level_gen_tuning.gd	86	LevelGenTuning	Resource	Room and map generation. Edit content/tuning/level_gen_tuning.tres.
data/tuning/music_tuning.gd	12	MusicTuning	Resource	How the music reacts to play (issue #13). Edit content/tuning/music_tuning.tres.
data/tuning/player_movement_tuning.gd	35	PlayerMovementTuning	Resource	Player movement values added after the prototype (momentum, launchers, on-beat movement). Edit content/tuning/player_movement_tuning.tres.

File (under src/)	Lines	Class	Extends	Description
data/tuning/timing_tuning.gd	35	TimingTuning	Resource	How attack timing is graded and rewarded (issues #12 and #14). Edit content/tuning/timing_tuning.tres.
data/tuning/timpani_tuning.gd	52	TimpaniTuning	Resource	Pattern and speed for the Hollow Timpani (issue #27: slowed down). Edit content/tuning/bosses/timpani_tuning.tres.
debug/debug_menu.gd	118	DebugMenu	Overlay	Testing shortcuts (issue #21). Opened with F1 inside a room; see DebugTools.
debug/debug_tools.gd	20	DebugTools		Whether the debug menu (issue #21) is available, and its key. Available in debug builds, or in any build launched with --debug-menu (godot ... -- --debug-menu).
fx/effects/base.gd	15	FxBase	Node2D	
fx/effects/column.gd	56	FxColumn	FxBase	A column that lights up, then strikes (the harp's strings, the staff's lines).
fx/effects/damage_number.gd	23	FxDamageNumber	FxFloatText	A damage number that can grow: later hits in the same window add into it.
fx/effects/decoy.gd	24	FxDecoy	FxBase	A decoy note (Ghost Note). Enemies target it; it pops.
fx/effects/float_text.gd	20	FxFloatText	FxBase	
fx/effects/pickup.gd	41	FxPickup	FxBase	A sharp or a staccato-dot heal. Bursts out, then homes to the player.
fx/effects/pillar.gd	49	FxPillar	StaticBody2D	A stone pillar raised from the ground. It is solid while it stands.
fx/effects/ring.gd	36	FxRing	FxBase	An instant area hit with an expanding ring drawn afterwards.
fx/effects/shockwave.gd	47	FxShockwave	FxBase	A wave that runs along a surface. Jump it.
fx/effects/slash.gd	26	FxSlash	FxBase	A crescent swipe.
fx/effects/spawn_mark.gd	19	FxSpawnMark	FxBase	The ink blot that becomes an enemy.
fx/effects/splat.gd	23	FxSplat	FxBase	Ink droplets that burst from a hit or a death.
fx/effects/tornado.gd	31	FxTornado	FxBase	
fx/effects/trail.gd	26	FxTrail	FxBase	A slipstream left by a dash (wind set bonus).
fx/fx.gd	18	FX		Short-lived things that live in a room: hit effects, area attacks, pickups, hazards. Each is a Node2D whose `room` is set by Room.add_fx(). Each effect lives in src/fx/effects/. These constants keep FX.Ring.new() and friends...
fx/projectile.gd	145	Projectile	Node2D	A shot, from either side. Collision is done by distance against the room's lists, which keeps it deterministic and cheap: a room never holds more than a few dozen of these.
input/input_bindings.gd	160	InputBindings		Every rebindable action, its defaults, and saving/loading the player's own bindings (issues #16 and #17). Bindings are stored in Game.settings["bindings"] as {action: [event, ...]} using the small dictionaries from to_dict().
main.gd	54		Node	Holds whichever screen is showing and fades between them.
ui/choice.gd	106		Overlay	Pick one of several items, laid out as cards. Mouse, keys and controller.
ui/controls_menu.gd	162	ControlsMenu	Overlay	Rebind every action (issues #16 and #17): three keyboard/mouse slots and one controller slot per action. Pick a slot, press the new input. Delete clears it.
ui/dialogue.gd	42		Overlay	Lines of speech, one at a time, typed out.
ui/ending.gd	81		Control	After a climb: fallen (Tacet), the first ending (Prima volta), or the secret one (Coda).
ui/hud/hud.gd	66	Hud	CanvasLayer	The HUD root. Builds one widget per job and forwards the room's requests to them. Widgets only draw; overlays (menus, choices) are added as children of this layer.

File (under src/)	Lines	Class	Extends	Description
ui/hud/hud_announcer.gd	39	HudAnnouncer	HudWidget	The big centered banner for room titles, boss names and milestones.
ui/hud/hud_boss_bar.gd	21	HudBossBar	HudWidget	Top-center boss name and health, with a tick at each phase change.
ui/hud/hud_character_card.gd	45	HudCharacterCard	HudWidget	Picking a note at the statues slides a fixed-width card in from the left edge, instead of a full-screen banner whose size depended on the text (issue #5).
ui/hud/hud_edge_flash.gd	31	HudEdgeFlash	HudWidget	A soft colored vignette when the player is hurt.
ui/hud/hud_health_panel.gd	35	HudHealthPanel	HudWidget	Top-left: the note, health as an inked staff, sharps, clefs, shield.
ui/hud/hud_metronome.gd	48	HudMetronome	HudWidget	Four beats of the bar at the bottom of the screen. The lit dot is now; a gold ring flashes when an attack lands on the beat.
ui/hud/hud_power_bar.gd	47	HudPowerBar	HudWidget	Power slots and the dash pip, tucked under the health bar in the top-left corner so they don't cover the platforming (issue #19).
ui/hud/hud_power_icons.gd	20	HudPowerIcons		Drawn icon for each power family, shared by the power bar and menus.
ui/hud/hud_prompt.gd	30	HudPrompt	HudWidget	The interact prompt above the metronome. For shop items it also shows what the item does (issue #20).
ui/hud/hud_room_status.gd	26	HudRoomStatus	HudWidget	Top-right: which bar you're in and what the room still wants from you.
ui/hud/hud_toasts.gd	37	HudToasts	HudWidget	Short notices stacked on the right: items gained, powers learned, enemy tips.
ui/hud/hud_widget.gd	27	HudWidget	Control	Base for one piece of the HUD. Fills the screen, ignores the mouse and redraws every frame from what it reads. A widget never changes game state.
ui/input_labels.gd	50	InputLabels		Short, readable names for whatever is bound to an action, so on-screen hints stay right after rebinding.
ui/loadout.gd	174		Overlay	Your note's whole kit: powers, rune slots, the rune bag, permanent and margin runes, relics, and which family sets (and dissonances) are awake. Runes can only be swapped outside combat.
ui/map/map_note_glyphs.gd	64	MapNoteGlyphs		One engraved note per room type, so the map reads like a score (issue #24). fight: quarter · elite: half + accent · shop: sharp + quarter · treasure: whole teacher: eighth · rest: half under a fermata · boss: breve
ui/map/map_sheet_renderer.gd	121	MapSheetRenderer		Draws a bar's map as a page of sheet music (issue #24): a titled staff with clef and time signature, one numbered measure per layer of rooms, rooms as notes, paths as slurs, cleared measures inked in the bar's colour and a...
ui/map_screen.gd	129		Control	A page of the climb, drawn as a measure: each layer a beat, each room a note on the staff. Choose the next note to play.
ui/menu_list.gd	58		Overlay	A vertical list of options. callback(index) or callback(-1) on cancel.
ui/overlay.gd	71	Overlay	Control	Base for anything that pauses the room and takes input: choices, dialogue, menus. Input is read by polling actions so keyboard, controller and mouse all work the same.
ui/pause_menu.gd	76	PauseMenu	Overlay	Pause: resume, settings, controls, give up / back to title, quit.
ui/settings_menu.gd	141	SettingsMenu	Overlay	Settings with sliders for every percentage and offset (issue #18). Drag with the mouse, or use left/right on a keyboard or controller.
ui/title.gd	159		Control	Title: the Grand Score above, a quarter note falling off it.

File (under src/)	Lines	Class	Extends	Description
ui/ui.gd	87	UI		Drawing helpers shared by every screen: text, panels, cards.
world/background.gd	141		Node2D	The page itself: paper, the faint staff that runs under everything, bar lines, the clef at the start, watercolour once the room is freed, and the inked segments you stand on.
world/events.gd	214	Events		What happens when you press interact on something.
world/generation/bar_planner.gd	63	BarPlanner		Lays out the rooms of a pillar bar (issue #8: longer bars, more power-ups). Layers: one entry fight, the branching layers from tuning, a fermata (sometimes with a chest beside it), then the keeper.
world/generation/feature_placer.gd	29	FeaturePlacer		Picks x positions for floor features so they never overlap (issue #4) and stay clear of the spawn and the exit.
world/generation/platform_reachability.gd	66	PlatformReachability		Makes sure every generated platform can be reached from the floor (issue #7). Walks up from the floor with the most limited jump (one line at a time), and gives any platform nobody can reach a stepping-stone ledge on the line...
world/generation/spawn_picker.gd	78	SpawnPicker		Chooses where an enemy appears: spread over the floor, every platform and the air, away from the player and from the other enemies in the same wave (issue #25).
world/generation/wave_plan.gd	12	WavePlan	RefCounted	One wave of a fight: who spawns at the start, and who may arrive as reinforcements once half of the wave is down.
world/generation/wave_planner.gd	53	WavePlanner		Plans the waves for a fight room with more variety (issue #25): a varying number of waves and sizes, a mix of enemy types with a cap on repeats, and occasional reinforcements.
world/interactable.gd	194	Interactable	Node2D	Anything the player can walk up to and press interact on: chests, shop pedestals, teachers, benches, character statues, signs, and the Scribble in the margin.
world/layout.gd	162	Layout		Where the ink goes. Each room gets a floor, walls, and the five staff lines broken into segments you stand on; each pillar shapes those segments differently and adds its own features (drums, updrafts, harmonic nodes).
world/map_gen.gd	68	MapGen		A page is a small branching graph drawn as a measure of music: each layer is a beat, each node a note you can play next. Types: combat, elite, shop, chest, teach, rest, boss One combat room per pillar page secretly hides the...
world/room/damage_numbers.gd	60	DamageNumbers	RefCounted	Turns every hit into on-screen numbers without stacking duplicates. Hits on one target within the merge window add into a single number per channel, so a swing plus its follow-ups (Sustain, Double Stop, crashes) reads as one hit.
world/room/room.gd	245	Room	RoomFlow	One room of a page. Builds its geometry, runs its waves or its event, holds the lists everything else queries, and hands out the rewards.
world/room/room_combat.gd	139	RoomCombat	RoomState	Layer 2 of 4. Waves, spawning, the enemy lists, the Fermata freeze.
world/room/room_flow.gd	140	RoomFlow	RoomCombat	Layer 3 of 4. What happens when things end: kills, clears, rewards, boss defeats, death, leaving the room.
world/room/room_state.gd	220	RoomState	Node2D	Layer 1 of 4. Every field a room has, geometry, and the helpers everything calls (fx, text, shake, announce, timers, overlays). Split from the original room.gd without logic changes.

7. Design decisions

Decision	Why	Trade-off / note
All art and audio made in code	No asset pipeline needed for a prototype; the ink-on-paper look is consistent everywhere.	Replacing it with sprites means rewriting each <code>_draw()</code> .
One scene, nodes built in code	Fast to generate and change; nothing hidden in the editor.	The editor shows almost nothing; the rebuild moves to scenes with exported references.
Staff lines are the platforms	Makes 'the map is a music sheet' literal; the climb is the core movement.	Line spacing (110 px) ties level design to jump height.
One global beat clock	Enemies, bosses, music and grading share one rhythm, so listening helps.	The clock runs while paused; listeners must check pause.
The Rest is audible silence	Hushed rooms are low-passed and lose the melody; clearing restores it.	Now graduated by how much of the room is left (#13).
Hades-style branching bars	Clear choices and pacing; room code is independent of the map.	Map style was TBD; a freezone or vertical layout could replace MapGen.
Areas called Bars in text only	Matches the music theme (#9) without renaming code.	Code still says page (page_id, PAGES).
Graded timing with a mash penalty	Rewards precision over spamming (#12).	Grades are judged at press time, not after the input buffer.
Eighth Note graded on half-beats	Half a beat is its note value; its fast attacks would otherwise mostly miss.	Only the Eighth; others use the beat.
Rhythm-pattern combos	Chosen by the owner (#11): the rhythm of your presses is the combo.	Two per character; data in .tres, easy to add more.
Full rebinding, no presets	Chosen by the owner (#17).	Controller stick bindings exist but the menu shows one controller column.
Super armor on elites and bosses	Stops stunlocking out of wind-ups (#3).	Hits still deal damage.
Detection range and patrol	Fixes wall-hugging (#2) and makes rooms feel less swarmed.	Enemies far away wait until you get close.
Reachability tested with Mundo	The most limited jumper; if Mundo can reach it everyone can (#7).	Stepping stones are added rather than platforms removed.
Tuning as .tres Resources	Designers edit numbers in the inspector; follows the rebuild rules.	Older prototype constants remain in code.
Layer chains for big classes	Every file under 300 lines without changing behaviour (#29).	A reading structure; the rebuild turns layers into components.
Seeded RNG everywhere in generation	The same run seed gives the same map; found and fixed an unseeded shuffle.	Combat randomness still uses the global RNG.
Test save isolation	Bots and captures once polluted the real save.	Anything run with <code>--script</code> uses a separate save file.

8. Playtest issue log

Issues from playtest round 1 (public repo JecerSE/discordant). Fixes are in the private repository only; the issues are still open on GitHub.

#	Issue	Status	Fix	Main files
1	Double damage numbers	Fixed	Hits on one target merge into one number per channel	damage_numbers.gd, damage_number.gd
2	Enemies hug the walls	Fixed	Detection range, patrol near spawn, turn cooldown	enemy_patrol.gd, enemy_move_ai.gd
3	Bosses stunlocked	Fixed	Super armor during wind-ups, stun immunity window	enemy_state.gd, enemy_damage.gd
4	Drum pads overlap	Fixed	Minimum spacing, clear of spawn and exit	feature_placer.gd
5	Jarring character select	Fixed	Fixed-width card slides in	hud_character_card.gd
6	Tempo changes	Deferred	Owner will revisit with the real music	
7	Unreachable platforms	Fixed	Reachability search with stepping stones	platform_reachability.gd
8	Longer bars, more power-ups	Fixed	6 branching layers, 2 chests, 2 teachers, bigger rooms, 40% bonus drop	bar_planner.gd, level_gen_tuning
9	Page → Bar	Fixed	Player-facing text	pages_data.gd, events.gd, ending.gd
10	Melee follows movement	Fixed	MeleeSwing child hitbox, down strike with pogo	melee_swing.gd, attack data
11	Combo system	Fixed	Rhythm-pattern combos with finishers	combo_tracker.gd, combo_finishers.gd
12	Reward precision	Fixed	Graded timing, mash penalty	beat_grader.gd
13	Fluid music	Fixed	Graduated muffle, crossfades	synth.gd, room_combat.gd
14	On-beat movement	Fixed	Jump, dash and landing bonuses	player_movement.gd
15	Momentum from launchers	Fixed	launch(), jump stacking, momentum carry	player_movement.gd
16	Rebinding and controller	Fixed	InputBindings, Controls menu	input_bindings.gd, controls_menu.gd
17	Control layout	Fixed	Full rebinding (owner's choice), no presets	same as #16
18	Sliders	Fixed	Settings menu with draggable sliders	settings_menu.gd
19	Power icons in the way	Fixed	Moved under the health bar	hud_power_bar.gd
20	Shop descriptions	Fixed	Prompt shows name, kind and effect	hud_prompt.gd, interactable.gd
21	God mode debug	Fixed	F1 debug menu	src/debug/
22	Mechanic in the wrong place	Deferred	Needs clarification from the owner	
24	Map as sheet music	Fixed	Score renderer and note glyphs	src/ui/map/
25	Randomised spawns	Fixed	Wave planner, spawn picker, reinforcements	wave_planner.gd, spawn_picker.gd
26	Relevant power-ups	Fixed	Requirement filter, pillar lean	item_requirements.gd, loot.gd
27	Slow the Percussion boss	Fixed	2-bar cycle, recovery, slower waves and mallets	timpani.gd, timpani_tuning
29	Split big scripts	Fixed	Folders and layer chains; powers per family	src/actors/player/powers/

#23 is the tracking issue and #28 does not exist.

9. Testing and tools

Tool	Command	What it checks
smoke.gd	godot --headless --path . --script tools/smoke.gd	Loads every script; a bot plays every room type on every bar with all items and powers, plus the secret boss
playthrough.gd	godot --headless --fixed-fps 60 --path . --script tools/playthrough.gd -- quarter 1	A full run through the real flow to an ending (1 = hold the clefs → secret ending); reports rooms that time out
test_generation.gd	godot --headless --path . --script tools/test_generation.gd	300 seeds: reachability, feature spacing, wave mix and caps, bar layout
test_combat.gd	godot --headless --path . --script tools/test_combat.gd	Grade thresholds, damage ordering, mash penalty, every combo pattern
test_menus.gd	godot --headless --path . --script tools/test_menus.gd	Pause → Settings sliders, Controls rebinding and conflicts, save round trip, debug actions
capture.gd	godot --path . --script tools/capture.gd --	Screenshots of every screen (opens a window)
gen_combat_tres.py	python3 tools/gen_combat_tres.py	Writes content/combat/*.tres from tables
build_docs.py	python3 tools/build_docs.py	Builds this document

Last verified at this commit: smoke test 0 errors; full playthroughs as Quarter and Eighth reached the Coda ending through 29 rooms with 0 errors; all three rule tests pass.

10. Known limitations and next steps

- Room is still a god object (about 300 direct room. calls); actors hold untyped room references.
- Most prototype content is still GDScript constants and string ids; a typo is a silent no-op.
- Only one scene file; no exported node references in prototype code.
- Combat randomness (crits, Kazoo, Out of Tune) uses the global RNG, not a seeded stream.
- Hits are distance checks; fast projectiles can pass through small targets.
- Controller stick bindings work but are not shown in the Controls menu (one controller column).
- Issue #6 (tempo) waits for the real music; issue #22 needs clarification.
- Feel and balance of the new systems (grades, combos, on-beat movement, longer bars) have not been hand-tested.
- The rebuild plan (components, event bus, headless core, command queue, integer maths) is in ARCHITECTURE.md Parts 2 and 3.

11. Repository and release workflow

Repository	Visibility	Use
JecerSE/the-discordant (~/the-discordant)	Private	All development. Every change is committed and pushed here.
JecerSE/discordant (~/discordant)	Public	Clean release snapshots only, when the owner calls a build a good release. No working notes, no assistant attribution.

Public releases carry Windows and Linux zips built from export_presets.cfg (templates for 4.7.2 in ~/local/share/godot/export_templates/4.7.2.stable).

12. Commit history

Commit	Date	Summary
bef1089	2026-09-25	Fix playtest issues #10 #11 #12 #13 #14 #16 #17 #18 #21 #29
ff8abdc	2026-09-25	Fix playtest issues #4 #7 #8 #24 #25 #26
e30aeb7	2026-09-25	Fix playtest issues #1 #2 #3 #5 #9 #15 #19 #20 #27

Commit	Date	Summary
b04a92d	2026-09-25	Split every script under 300 lines, grouped by system
e66dc36	2026-09-25	Stun formula note; editor re-save of project.godot
f784849	2026-09-25	Architecture document: add rebuild rules, reference player controller and migration map
b6e9c4f	2026-09-25	Add ARCHITECTURE.md: technical audit of the prototype for the rebuild
477aaed	2026-09-24	Plainer in-game text, export presets, simpler README
7b09da4	2026-09-24	Playthrough bot: guard freed hub screen
bf76210	2026-09-24	The Discordant v1: rhythm roguelite platformer on the Grand Score

13. Glossary

Term	Meaning
Bar	An area of the climb (Percussion, Wind, Strings, the Grand Score). Called page in code.
Keeper	A pillar bar's boss.
The Rest / hushed	The corruption; a room with enemies left is hushed.
Sharps (#)	Currency.
Fermata	A rest room (heal or rehearse), also a Rest power and a pause rune.
Rehearse	Raise a power's level at a fermata.
Family / pillar	Percussion, Wind or String. Rest (margin) is the fourth, secret family.
Dissonance	Rival pillars' family runes equipped together.
Grade	Perfect, great, good or miss: how close a press was to the beat.
Layer chain	A class split across files, each file extending the previous one.
Tuning resource	A Resource script plus a .tres file holding balance numbers.